

Sahil Niranjana

AI/ML Engineer

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🌐 LinkedIn

🌐 Portfolio Website

🌐 GitHub

SUMMARY

AI/ML Engineer with 3+ years of experience designing and deploying production-grade machine learning systems across the full model lifecycle. Combines strong engineering discipline with a research-driven mindset to build scalable pipelines, ship measurable improvements, and translate complex model outputs into tangible business value.

SKILLS

Programming Languages: Python, R, SQL

ML Frameworks & Algorithms: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Pandas, NumPy, CNNs, LSTMs, Gradient Boosting, Random Forest, Decision Trees, Linear & Logistic Regression

GenAI & LLMs: LangChain, LangGraph, OpenAI API, Hugging Face, Transformers, BERT, RAG Pipelines, Prompt Engineering, LLM Fine-Tuning

Data Engineering & MLOps: Apache Spark, Airflow, Apache Kafka, dbt, ETL Pipelines, Feature Engineering, MLflow, Weights & Biases (W&B), Docker, Kubernetes, GitHub Actions CI/CD

Cloud & Deployment: AWS SageMaker, Flask, Streamlit, REST APIs

Databases & Data Visualization: MySQL, PostgreSQL, MongoDB, Power BI, Tableau

Version Control: Git, GitHub

PROFESSIONAL EXPERIENCE

AI/ML Engineer | Bank of New York, Massachusetts

January 2026 – April 2026

- Developed scalable **Python** data pipelines on **AWS SageMaker** within a cross-functional **ML team** to automate feature engineering and streamline **model training** data ingestion, reducing end-to-end data retrieval latency by 44% under strict federal compliance standards.
- Built **agentic AI** multi-agent workflows using **LangChain** and **LangGraph**, integrating **RAG pipeline** for compliance document retrieval to automate treasury account verification with human-in-the-loop checkpoints, reducing manual processing by 35%.
- Deployed real-time anomaly detection models using **Docker** and **Kubernetes** into production backend systems, reducing fraudulent transaction exposure by 15% and cutting annual financial risk by \$2M.
- Engineered core **deep learning** model components using **PyTorch** for a proprietary **Snowflake**-integrated cash forecasting feature, increasing rolling prediction precision to 85% and helping enterprise clients effectively optimize short-term liquidity management.
- Built and maintained **MLflow** CI/CD automation pipelines supporting model retraining and redeployment of banking models, reducing engineering maintenance overhead by 25% with zero-downtime upgrades across distributed microservices infrastructure.

AI/ML Engineer | Vivma Software Inc, India

January 2022 - August 2024

- Developed ML-based financial anomaly detection and risk validation models using **XGBoost** and **ensemble methods** (gradient boosting, random forests), increasing risk-detection precision by 31% and reducing manual review workload across compliance workflows
- Engineered large-scale data pipelines processing 1.5M+ records monthly of financial data across 4 teams using **Python**, **SQL**, **Pandas**, and **Apache Spark**, orchestrated via **Apache Kafka** for real-time ingestion and fault-tolerant stream processing
- Managed Git-based **version control** and **CI/CD workflows** using **GitHub Actions** to automate model testing and deployment across shared **ML pipelines**, improving code integration consistency and reducing release cycle time across teams.
- Established model evaluation pipelines and validation frameworks using **Scikit-learn** metrics and **Weights & Biases (W&B)** for experiment tracking, reducing data discrepancies by 38% and enabling reproducible benchmarking across model iterations
- Standardized **SQL** and **dbt**-based feature extraction pipelines across 3 teams, improving data consistency by 25%, reducing processing time and delivering **\$850K in annual cost savings** that directly informed budget planning for the data infrastructure roadmap

PROJECTS

LOBPredictor- Latency-Optimized ML Inference Engine | [GitHub](#) | [Live Demo](#)

Constructed end-to-end ML pipeline predicting mid-price moves from L2 order-book data, engineering 18 features (order-flow imbalance, microprice, rolling windows) with Scikit-learn; designed evaluation/backtesting framework (Sharpe, drawdown, win rate) and deployed C++ inference engine at less than 0.7 ms p99 latency.

Customer Behavior Analytics Platform | [GitHub](#) | [Live Demo](#)

Developed a ML platform with 4 models- K-Means segmentation, XGBoost churn prediction (0.90+ AUC-ROC), pattern detection, and recommendation engine processing 50K+ transactions at 97% accuracy; deployed with Flask API, Streamlit, and SQL at 99% uptime.

EDUCATION

Masters of Professional Studies in Analytics | Northeastern University, Boston, USA

June 2026

Advanced Certificate Program in Data Science | International Institute of Information Technology, India

October 2023

Bachelor of Technology in Computer Science and Engineering | Guru Gobind Singh Indraprastha University (GGSIU), India

June 2022

PUBLICATIONS

Patent 202211030731 (India, Granted): Real-time 3D pose classification system using ML models achieving 97% accuracy. ([Link](#))

Published: Emotion classification via multi-modal deep learning- IJEAST, Vol. 6, No. 7. ([Link](#))